

Phase 0 Coverage & Feasibility Report

Project: AFA 2027 QTM Crypto. **Phase:** 0 (Asset Universe). **Generated by:** 04_code/coverage_report.py.

Sample: 2015-08-31 through 2026-05-31 (130 monthly snapshots, month-end frequency).

Primary data source: CoinMarketCap free historical listings data-api (point-in-time, includes delisted coins). See DATA_DECISIONS_LOG.md.

1. Headline counts

- **Qualifying assets ever (N=250, ex-stablecoins):** 1939
- **Live (observed in top-1000) cross-section:** ranges 245–632 assets/month.
- **Observed within top-250 each month:** ranges 217–248 (short of 250 because excluded stablecoins occupy real top-250 rank slots).
- **Class counts (by distinct asset):** coin 618, token 447, other 874.
- **Class weight (by *observed* asset-month, i.e. the live universe):** coin 26869, token 16540, other 23894.
- **Stablecoins excluded (hit top-250 at least once):** 71.

2. Coverage by year (as of each year's last snapshot)

Year	As of	Observed	Obs≤250	Coins	Tokens	Other	New entrants
2015	2015-12-31	288	245	196	19	73	312
2016	2016-12-31	347	243	256	17	74	152
2017	2017-12-31	582	247	336	54	192	285
2018	2018-12-31	531	243	228	68	235	198
2019	2019-12-31	564	239	213	83	268	164
2020	2020-12-31	627	235	222	139	266	202
2021	2021-12-31	538	234	174	179	185	165
2022	2022-12-31	572	229	180	201	191	58
2023	2023-12-31	547	236	170	205	172	85
2024	2024-12-31	535	237	164	198	173	140
2025	2025-12-31	581	221	159	233	189	147
2026	2026-05-31	540	217	146	207	187	31

3. Where the panel is thin, and what it gates

- **Governance tokens reach ≥ 10 observed names in 2015-08-31; ≥ 30 in 2017-07-31.** Before that, H1b / H2 / H3 (which need governance-token cross-sections and voting data) are not meaningfully testable — the early panel is coin-dominated.
- **Coins reach ≥ 30 observed names in 2015-08-31.** H1a (coins) has usable cross-sectional density from the early sample onward, though 2015–2016 is still thin and dominated by BTC/LTC/XRP-era assets and many later-failed alts.
- The 2015–2019 window is coin-heavy and token-sparse exactly as the spec anticipated; treat early-sample H1b/H2/H3 results as out-of-support.

4. The 'other' bucket and classification caveats

- 874 assets (45% of distinct names) are class='other' — oracle/utility/exchange tokens, wrapped/tokenized assets, memecoins, and the long tail of thin-tagged dead microcaps that briefly touched the top-250 and were retained per the anti-survivorship rule.
- Weighted by *observed asset-months* (the live universe), 'other' is a much smaller share than its raw name-count suggests; the persistent, higher-cap assets that drive the tests are predominantly classified coin/token.
- **Classification is a tag-driven first pass and is flagged for review.** CMC tags are *current* metadata, not point-in-time, so consensus/governance status is the asset's end-state. Time-varying cases are handled explicitly where known (ETH: staking from 2020-12-01, full PoS at the 2022-09-15 Merge) and flagged elsewhere. `ambiguous_flag=True` marks every asset needing manual confirmation (1132 in-universe).

5. Stablecoins (excluded from the universe, per project decision)

- 71 stablecoins qualified for the top-250 at least once and were dropped from the panel so they never enter downstream phases (their ~ 0 returns and degenerate λ would otherwise contaminate the cross-section).
- Ranking was computed *inclusive* of stablecoins so the top-250 cutoff reflects true market structure; stablecoins were removed only from the output panel. This is why the observed-top-250 count runs a few-to-low-tens below 250 in later years.
- Note the cut is functional: TerraUSD (UST) is excluded as a stablecoin, while LUNA/LUNC (the Terra *coin*) is retained — including its collapse, a deliberate anti-survivorship inclusion.

6. λ -channel feasibility (look-ahead to Phase 1)

- **Staking/locking channel:** available for PoS coins once each chain's staking launches; absent for pure-PoW coins (no staking) and for ETH before 2020-12. vote-escrow locked supply (veCRV/veBAL) available for the relevant tokens from ~ 2020 onward.
- **Voting-engagement channel:** essentially unavailable before ~ 2020 and absent for pure-coin assets; depends on Snapshot/Tally/on-chain governance coverage.
- **Holding-duration channel:** computable from on-chain data for major chains but needs per-chain explorer work; the only λ channel realistically available for early-sample coins.

7. Rank-threshold sensitivity (N)

Top-N	Qualifying total	Non-stablecoin	Stablecoins excluded
200	1612	1550	62
250	2010	1939	71
300	2354	2275	79

Default N=250 is used for the panel. The count of qualifying names scales roughly linearly with N; the live monthly cross-section is what matters for the tests and is reported above.

8. Pre-2015 native-coin extension (supplementary note)

Per spec, no asset's series is extended before 2015-08 in the main panel (common start for comparability). For a possible coins-only robustness subsample later (paper Section 5), pre-2015 history exists chiefly for **BTC** (genesis Jan 2009; CMC price history from $\sim 2013-04$), **LTC** (2013), **XRP** (2013), **DOGE** (2013), and a handful of other

2013–2014 coins. CMC's historical data-api serves snapshots back to ~2013-04-28, so a BTC-led coins-only sample could be extended ~2.3 years earlier if needed. This is noted, not built, in Phase 0.

9. Open questions for review before Phase 1

1. **Post-delisting months / carry-forward:** assets that fall below rank 1000 are marked `status='carried_forward'` with last-observed price retained (not dropped). Phase 3 must decide the death-return treatment (final delisting return vs. constant carry). Flagged, not resolved, in Phase 0.
2. **'other'/ambiguous classifications** need a manual confirmation pass for any asset that will actually enter the tests (i.e. observed, reasonably ranked).
3. **Confirm N=250** vs. a tighter screen if the live cross-section is judged too wide once λ -channel coverage is known.
4. **Artemis** access (paid/auth) if its on-chain fundamentals are wanted for PQ in Phase 2 — no free API available as of this build.